

Basic Principles of Medicine 2 (BPM501)

Module: Digestion and Metabolism

DLA: Digestion and Absorption

Dr. Felicia Ikolo

fikolo@sgu.edu

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Acknowledgement
Dr. Margit Trotz
Dr. Sharmila Upadhya

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Recommended Reading

- Lippincott Illustrated Reviews 9th Edition:
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Chapter 7 – Introduction to Carbohydrate: [Introduction to Carbohydrates | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#) Section: Dietary Carbohydrate digestion
- Chapter 15 - Dietary Lipid Metabolism: [15. Dietary Lipid Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Chapter 19 - Amino Acids: Nitrogen Disposal: [19. Amino Acids: Nitrogen Disposal | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#) : Section: Dietary Protein Digestion.

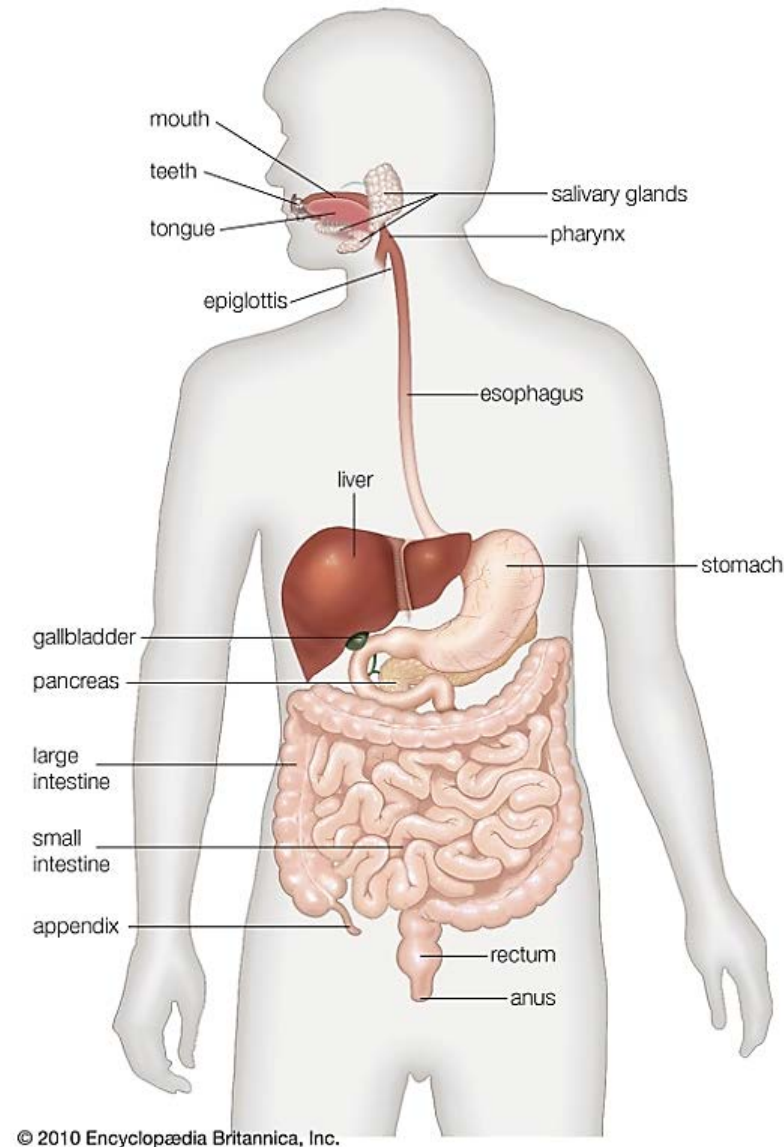
Objectives: DLA: Digestion and Absorption

- Before DM Lecture 28

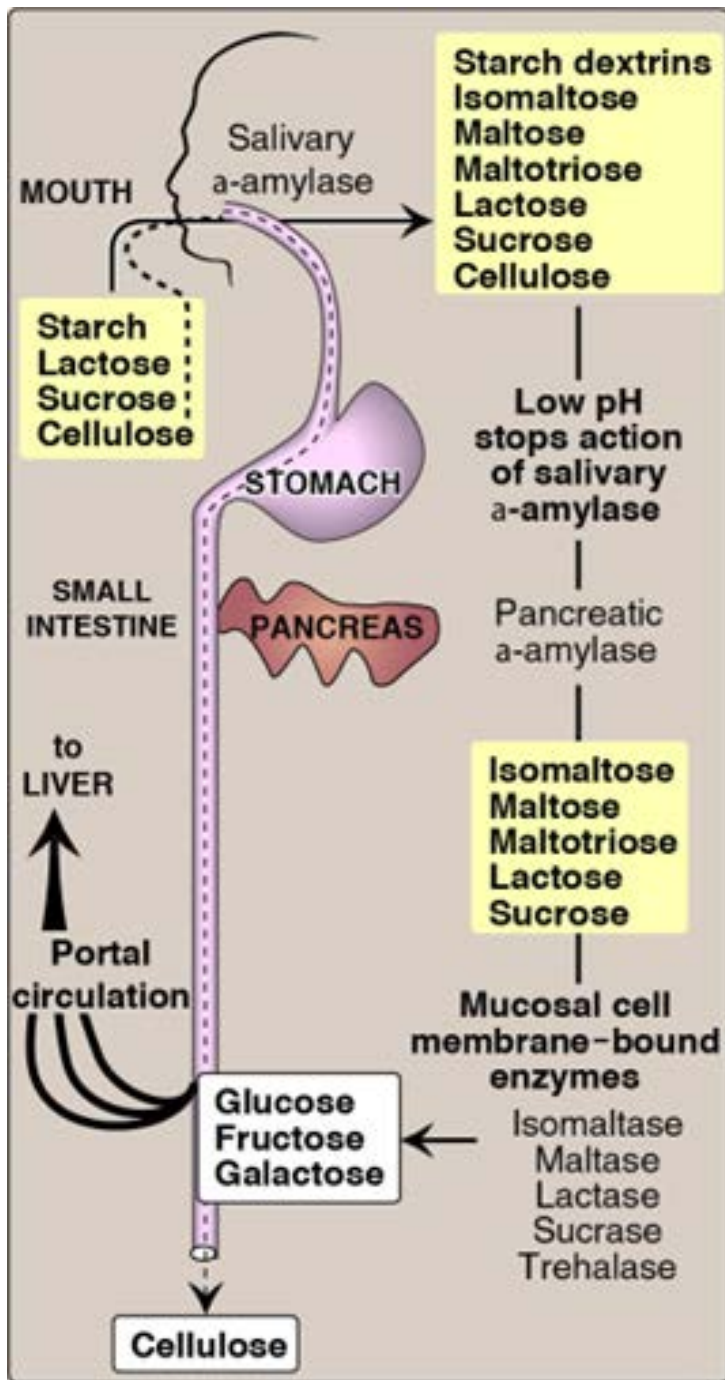
SOM. MK.I.BPM2.3.DM.3.BCHM.1237	Describe digestion in mouth and stomach (role of salivary amylase and gastric lipase).
SOM. MK.I.BPM2.3.DM.3.BCHM.1238	Describe role of gastrin and activation of pepsinogen in stomach. Discuss consequences of insufficient acid production in stomach and the effect of antacids. Discuss gastric digestion of milk TAGs.
SOM. MK.I.BPM2.3.DM.3.BCHM.1239	Explain mechanisms involved in release of cholecystokinin and secretin. Review functions of cholecystokinin and secretin. Indicate role of HCO ₃ ⁻ in pancreatic secretion and bile.
SOM. MK.I.BPM2.3.DM.3.BCHM.1240	Outline digestion and absorption of carbohydrates. Describe reactions of disaccharidases in brush border of intestinal cells.
SOM. MK.I.BPM2.3.DM.3.BCHM.1241	Describe lactose intolerance and differentiate primary, secondary and congenital lactose intolerance based on age of onset and primary defect.
SOM. MK.I.BPM2.3.DM.3.BCHM.1242	Outline lipid digestion and absorption and describe the significance of bile.
SOM. MK.I.BPM2.3.DM.3.BCHM.1243	Discuss differences between primary and secondary bile acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1244	Outline role of liver in synthesis of primary bile acids from cholesterol. Review formation of bile salts and their release into bile.

Digestion and Absorption

- **Digestion** is the process of breaking down complex food material into smaller and simpler forms in the digestive tract.
 - Includes activities in the alimentary tract and associated glands
 - Mechanical, chemical and microbial actions essential for **digestion** into simpler forms.
- **Absorption** is the passage of digested nutrients through the intestinal mucous membrane into the blood stream and into the cells.
- Carbohydrates, proteins, and lipids are **ingested** from a meal, and subsequently **digested** and **absorbed**



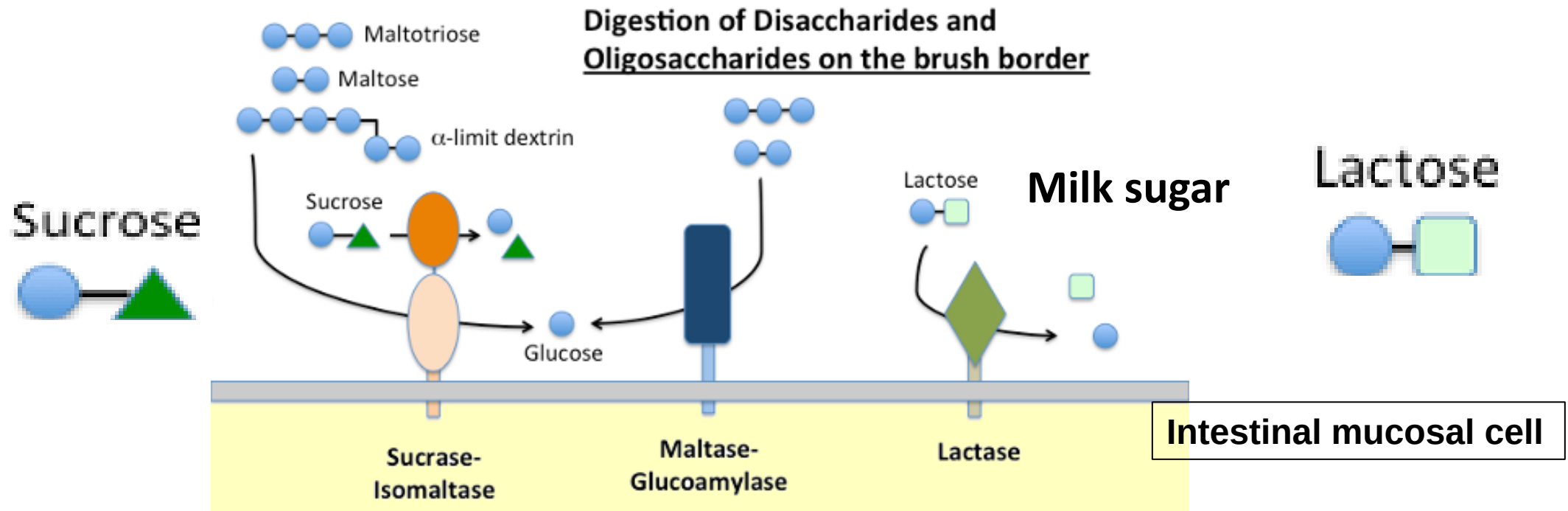
Overview of Digestion and Absorption of Carbohydrates.



- Carbohydrates digestion is mainly in the mouth and intestinal lumen.
- The final products of carbohydrate digestion, the monosaccharides (glucose, fructose and galactose) are absorbed by cells (enterocytes) of the small intestine.

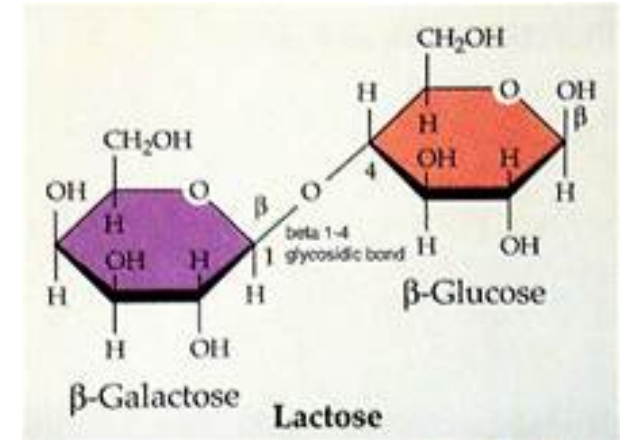
Disaccharidases

Disaccharidases on Intestinal Mucosal Brush Border



Sucrose, maltose and lactose: **NOT** digested by pancreatic or salivary amylase

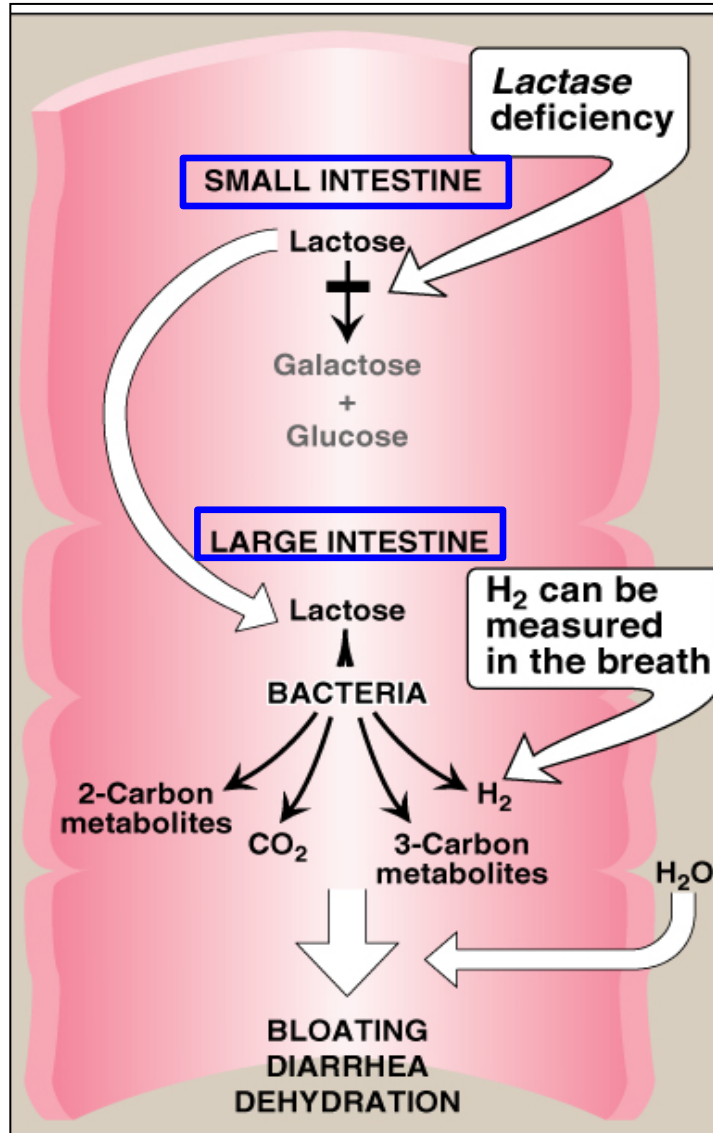
Lactose Intolerance



Lactose (milk sugar) is cleaved by lactase in the small intestine to glucose and galactose.

Deficiency of lactase leads to pain, nausea, flatulence and diarrhea after ingestion of food containing lactose.

Lactase Deficiency: Osmotic Diarrhea



Lippincott page 97, Fig. 7.11

- **Lactase deficiency** leads to undigested **lactose** reaching the large intestine.
- **Large intestine** has **bacteria (normal flora)** which degrade lactose to **osmotically active compounds** and in this process form **hydrogen gas**. (Abdominal cramps, bloating, flatulence)
- **Osmotic diarrhea: Undigested lactose (osmotically active) retains water in the lumen → Watery stools (diarrhea)**
 - **No reducing sugar in urine**
- **Lab test: Breath Hydrogen test – used for diagnosis**

Lactose Intolerance

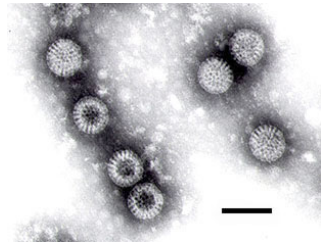
1. **Primary lactose intolerance:** (normal result of aging) *also known as* **lactase nonpersistence** or **adult hypolactasia**
2. **Secondary lactose intolerance:** (due to illness or injury) *name changed to* **disaccharidase deficiency**
3. **Congenital lactase deficiency:** (born with the condition) **Most severe** form and very rare.

1. Primary Lactose Intolerance

- Age dependent loss of lactase activity
- The levels of **lactase** is strongly reduced by about **seven years of age** and stays reduced.
 - Sometimes may develop during adulthood.
- Over 60% of adults (genetic variety) are lactose intolerant.
- **Primary lactose intolerance** is **very common** in adults of African or Asian descent. Less common in individuals of Northern European origin.

2. Secondary Lactose Intolerance

- **Transient:** All disaccharidases affected but **lactase is usually first enzyme lost and the last to recover.**
- Can occur due to **loss or damage of intestinal mucosa cells** and loss of villi and **brush border disaccharidases** (Food poisoning).
- **Severe diarrhea or gastroenteritis** due to **rotavirus** can lead to **intestinal injury**. Mainly in children and is contracted by the fecal-oral route.
- **Celiac disease** is **immune-mediated** causing **damage** to the intestinal **mucosa** in response to **gluten (protein in grains)**.



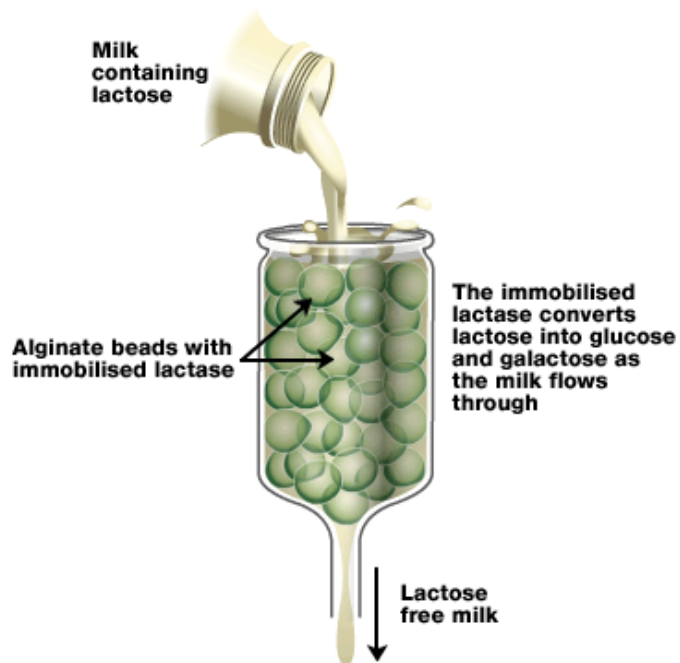
3. Congenital Lactase Deficiency

- **Generally, Lactase** activity is **highest** after **birth** for **digestion of milk lactose** (main nutrient)
- **Congenital lactase deficiency** is rare
- Leads to **severe osmotic diarrhea** and life-threatening **dehydration**
- The **baby cries** due to **painful bloating** (often **mistaken for hunger**)
- **Breast feeding** continues to cause **pain** and **diarrhea**.
- Management:
 - **Rehydration** and **STOP breast-feeding**
 - **Lactose-free infant formula**

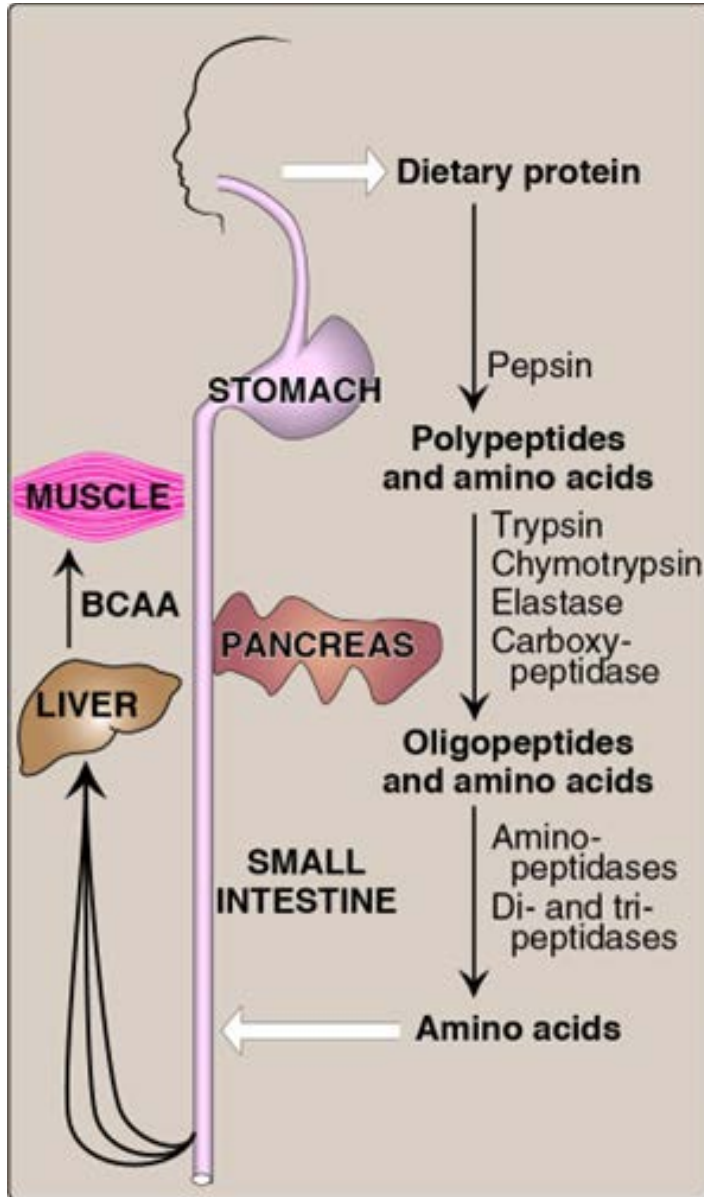


Management of Lactose Intolerance/Malabsorption

- Diet low in lactose and should **not** include milk and ice cream other than lactose-free products
- Some aged cheese or yogurt may be tolerated
- Lactose-free products and lactase pills available



Overview of Digestion and Absorption of Proteins.



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BCAA = branched chain amino acids.

- Proteins are too large to be absorbed in the intestine. An exception: newborns can take up maternal antibodies in breast milk.
- Proteins must be digested (hydrolyzed) to yield dipeptides, tripeptides, and individual amino acids, which can be absorbed.
- Proteolytic enzymes (digest proteins) are produced in the stomach, the pancreas, and the small intestine.

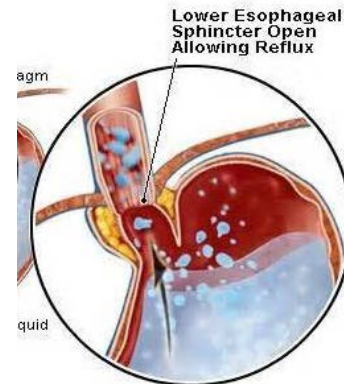
Hydrochloric acid (strong acid)

- **HCl denatures food** for quicker digestion and destroys **bacteria** and **fungi** present in food.
- HCl is also known as gastric acid.
- **HCl** and a high proton concentration **alters** the **conformation** of **pepsinogen** and facilitates **autocatalytic activation** of the zymogen to the **active protease pepsin**.
 - **Pepsin** has a pH 2 optimum; Cleaves **peptide bonds** within dietary **proteins (endopeptidase)** in stomach lumen.

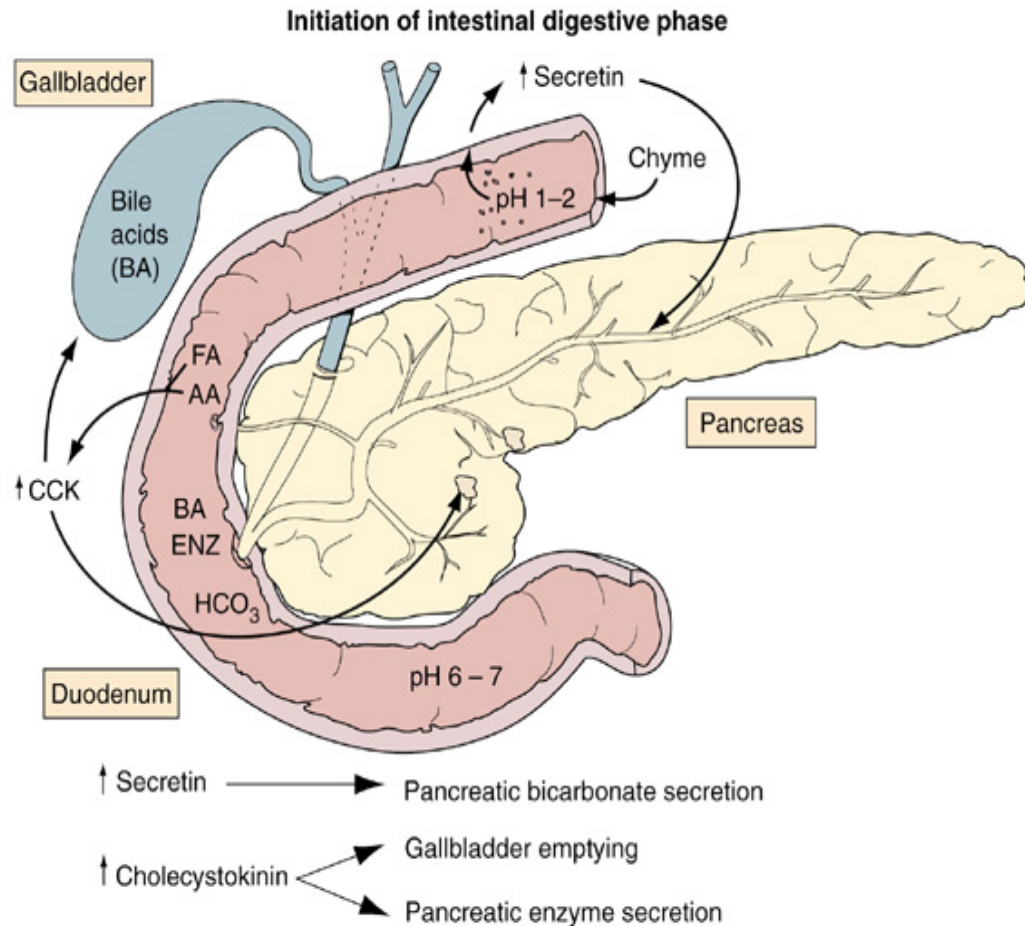
Achlorhydria

- **Achlorhydria (low gastric HCl)**

- Common in **elderly** due to **decline of HCl** production with age
- **Pernicious anemia** (due to autoimmune destruction of parietal cells)
- May result from **excessive ingestion of antacids** (used for acid reflux)



Release of Secretin and Cholecystokinin (CCK) from Endocrine Cells of the Duodenum

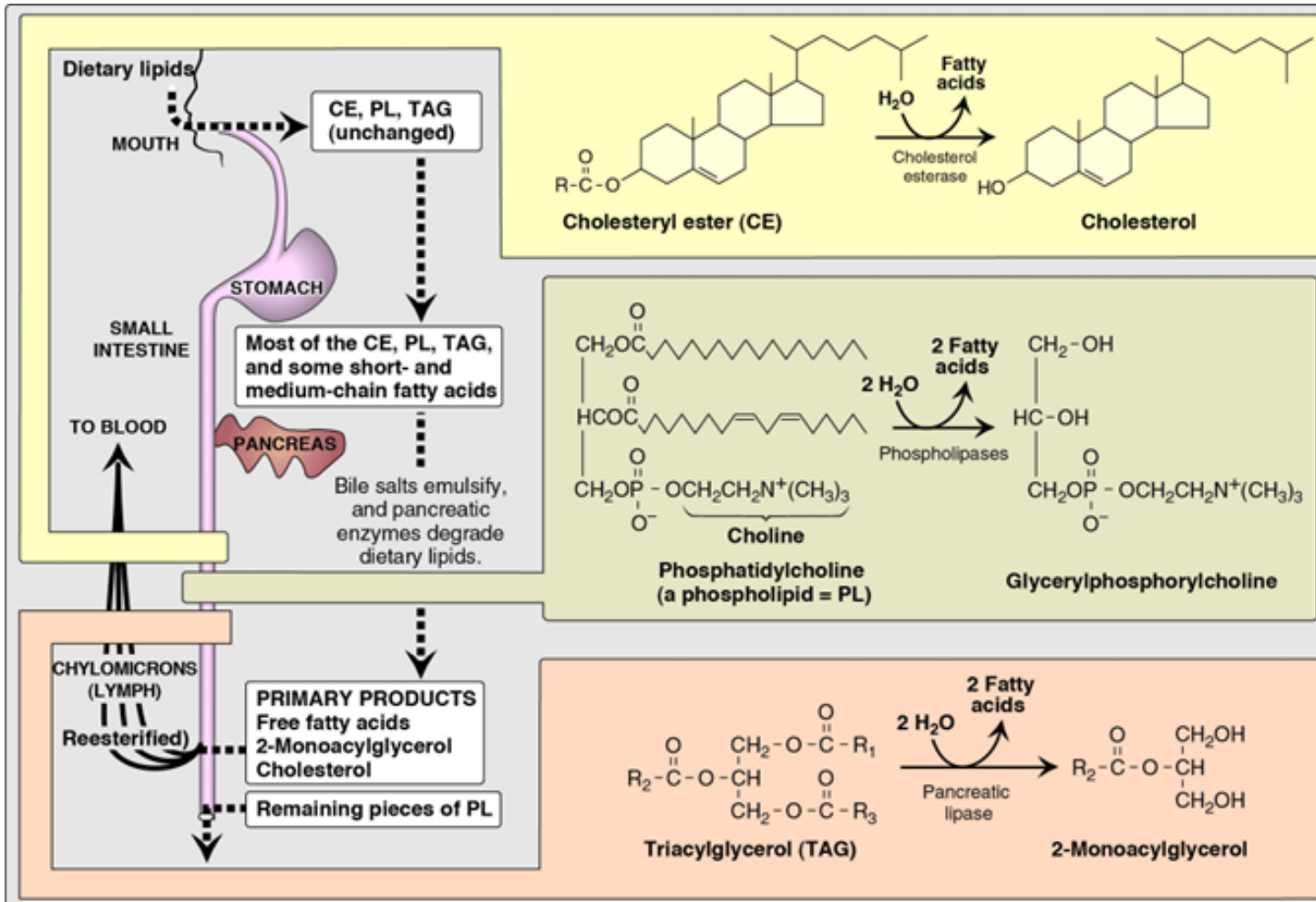


1. Acidic chyme reaches duodenum stimulates release of **secretin** which stimulates **pancreatic bicarbonate secretion**.
2. The pH is changed to a **neutral pH** which is needed for pancreatic enzymes.
3. Partially digested food due to digestion in stomach results in release of **CCK**.

Actions of Cholecystikin

- **Activates:**
 - **Release of bile** from **gallbladder**: Bile required for lipid digestion and absorption. Bile is also alkaline and neutralizes pH.
 - **Enzyme enteropeptidase** in the **duodenum** **activates** trypsinogen to **trypsin**.
 - **Secretion** of **pancreatic enzymes, zymogens** and other proteins.
- **Inhibits:**
 - **Gastric motility** and **food release** into the duodenum.

Overview of Digestion and Absorption of Lipids.

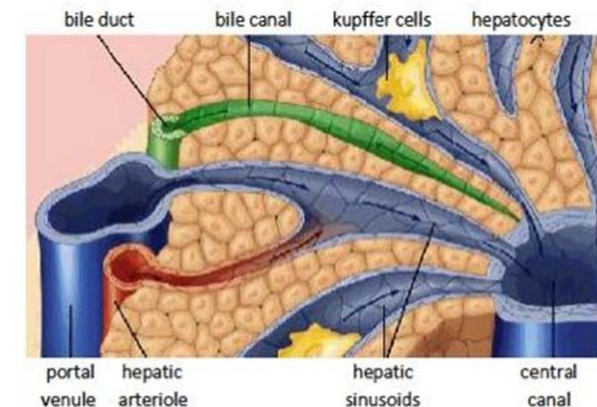
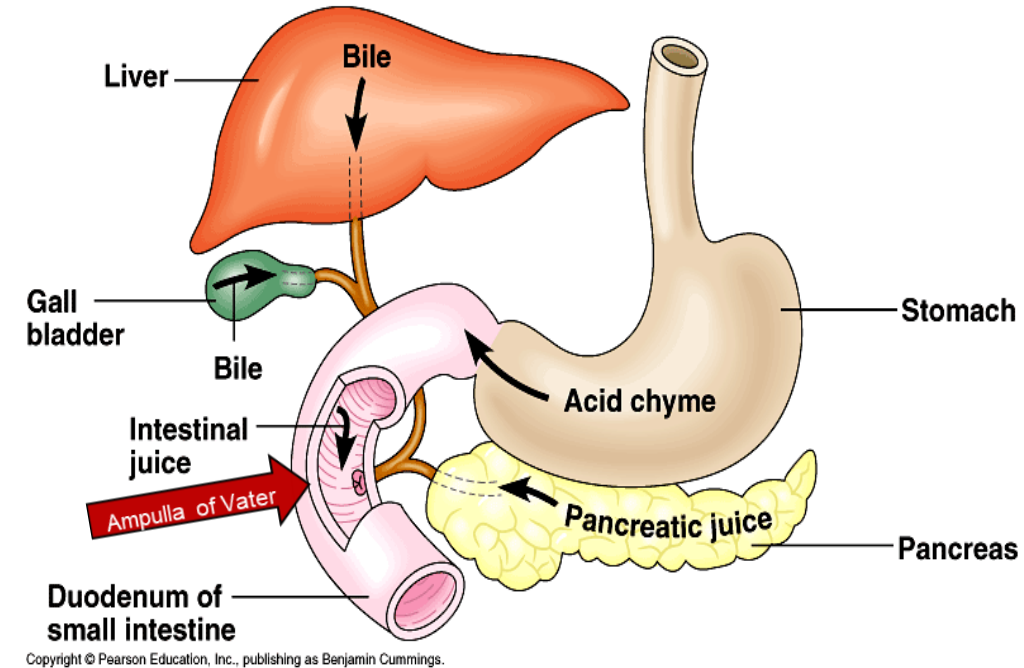


- Dietary lipid digestion begins in stomach
- Major site of lipid digestion: **Small intestine.**
- End-products of digestion: **Free fatty acids, free cholesterol, and 2-MAG** formed in the jejunum. These, with bile salts and fat-soluble vitamins (A, D, E, and K), form **mixed micelles**
- **Mixed micelles** transport the end products of lipid digestion to the brush border of the enterocytes.

- Bile salts are absorbed in the terminal ileum, with <5% being lost in the feces.
- Short- and medium-chain FA are water soluble (do not require mixed micelles for absorption)

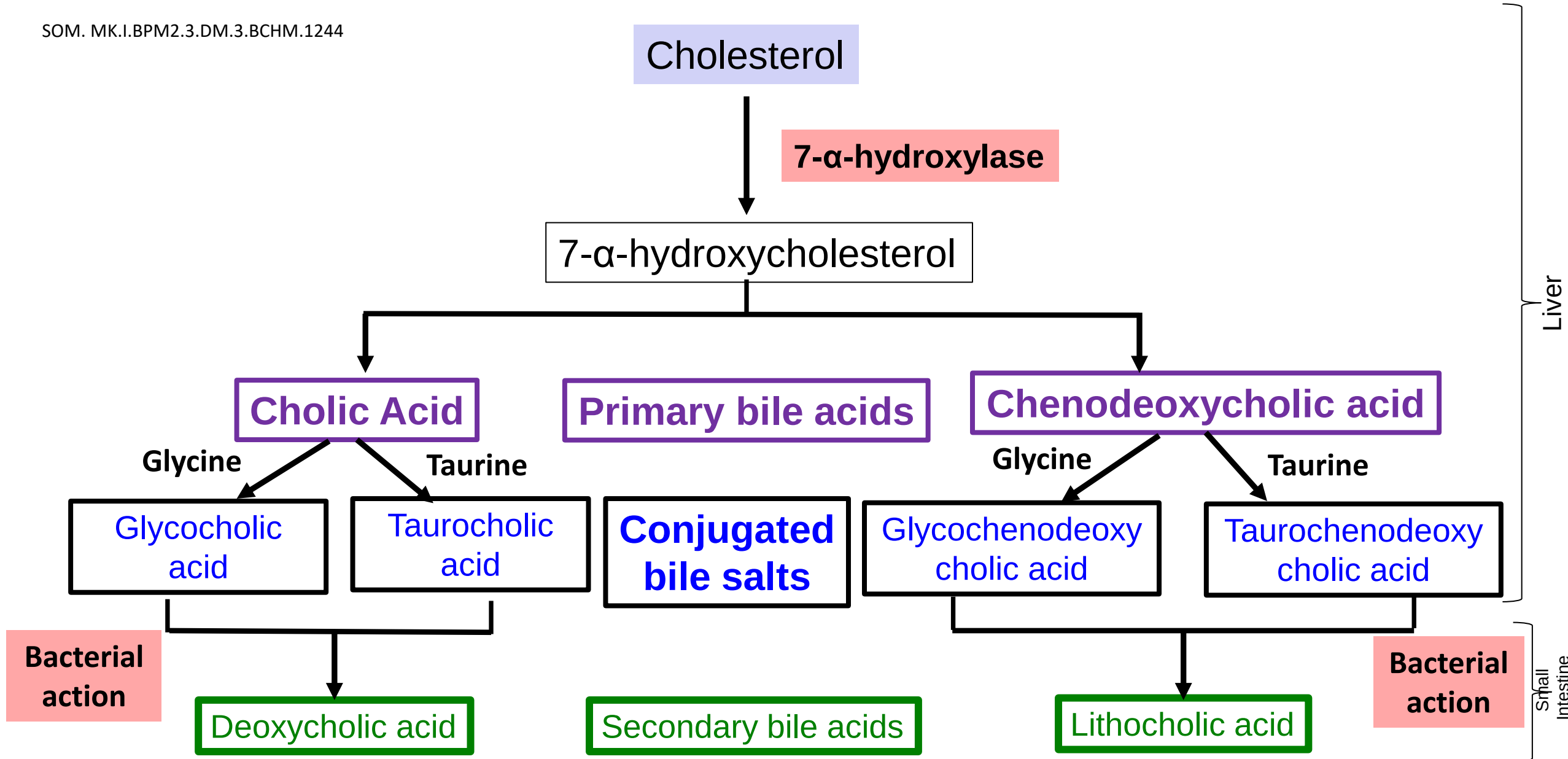
Bile: Composition and Functions

- **Alkaline fluid** formed and released by hepatocytes (liver) into the bile ducts
- Bile contains **conjugated bile salts** and phosphatidylcholine
 - Bile salts produced in liver and secreted into the bile.
 - **Bile is stored and concentrated in gall bladder** and released into the duodenum during a meal (CCK)
 - Bile salts act as detergents aiding in lipid digestion
 - Small amounts of **cholesterol** in bile (higher levels of cholesterol in bile favors formation of gall stones)
 - Bilirubin (bile pigment) gives yellow color

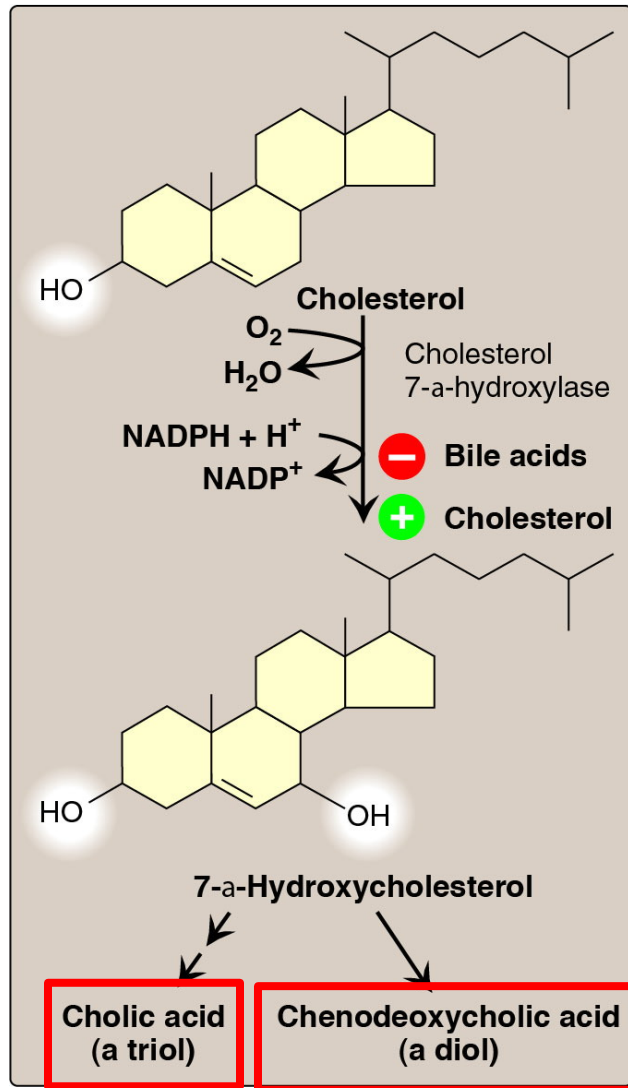


Overview: Metabolism of Bile Acids/Bile Salts

SOM. MK.I.BPM2.3.DM.3.BCHM.1244



Primary Bile Acids/Bile Salts (Used interchangeably)



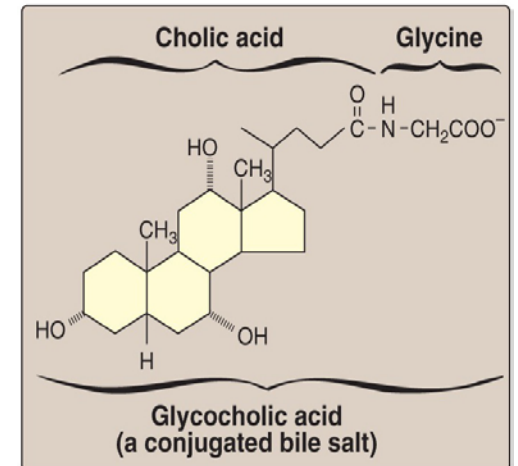
- Liver synthesizes bile acids from free cholesterol : Primary bile acids
- Primary bile acids: **Cholic acid and chenodeoxycholic acid**
- Regulated enzyme: **Cholesterol 7 α -hydroxylase**
- **Primary bile acids** are **conjugated** with glycine or taurine in **liver** to form **Conjugated** (primary) bile salts

HA = Protonated $\rightarrow$ bile acids (Glycocholic acid)
A $^-$ = Deprotonated (ionized) $\rightarrow$ bile salts (Glycocholate)

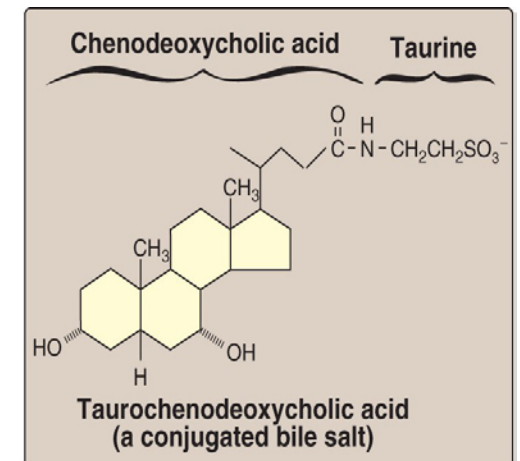
Primary bile acids

Conjugated Bile acid/salt: Synthesis in Liver

- **Primary bile acids** are **conjugated with glycine or taurine** in **liver** to form **Conjugated (primary) bile salts**
 - **Glycocholic acid and taurocholic acid**
 - **Glychenodeoxycholic acid and taurochenodeoxycholic acid**
- Conjugated bile salts (primary bile salts) are negatively charged in the small intestine (>pH 7)
 - Increase efficiency of dietary fat emulsification
- **Conjugated bile salts** released into bile and emulsify **dietary lipids** in the **duodenum, jejunum and ileum**.

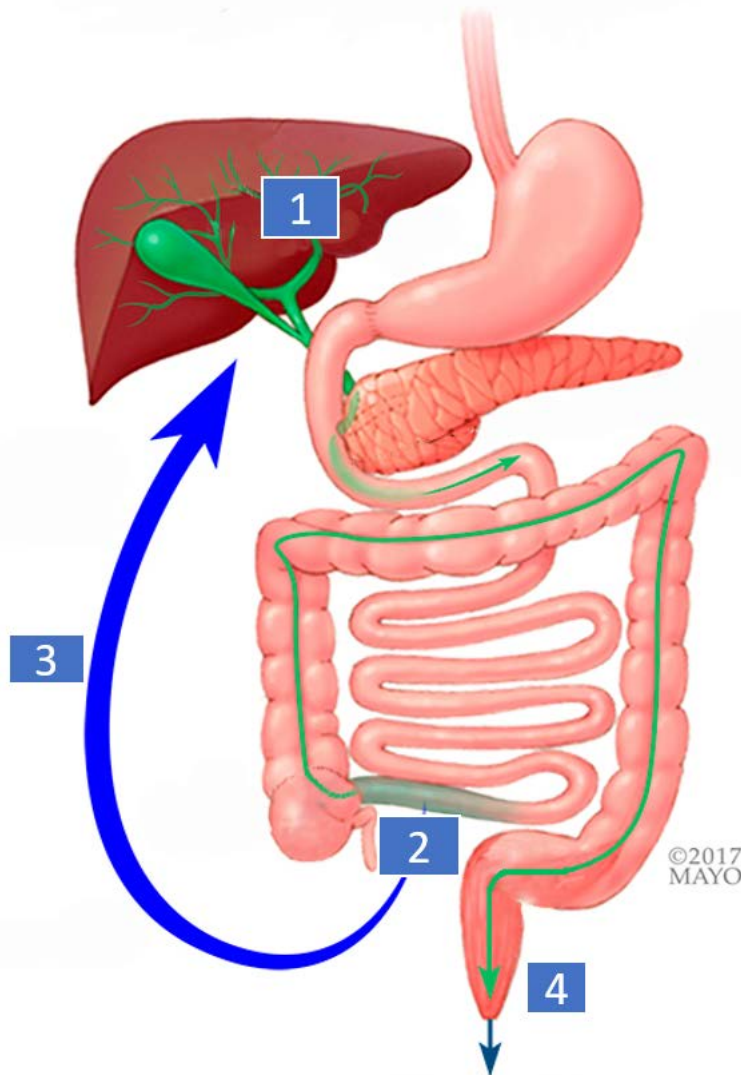


Or Taurocholic acid
(Cholic acid + taurine)



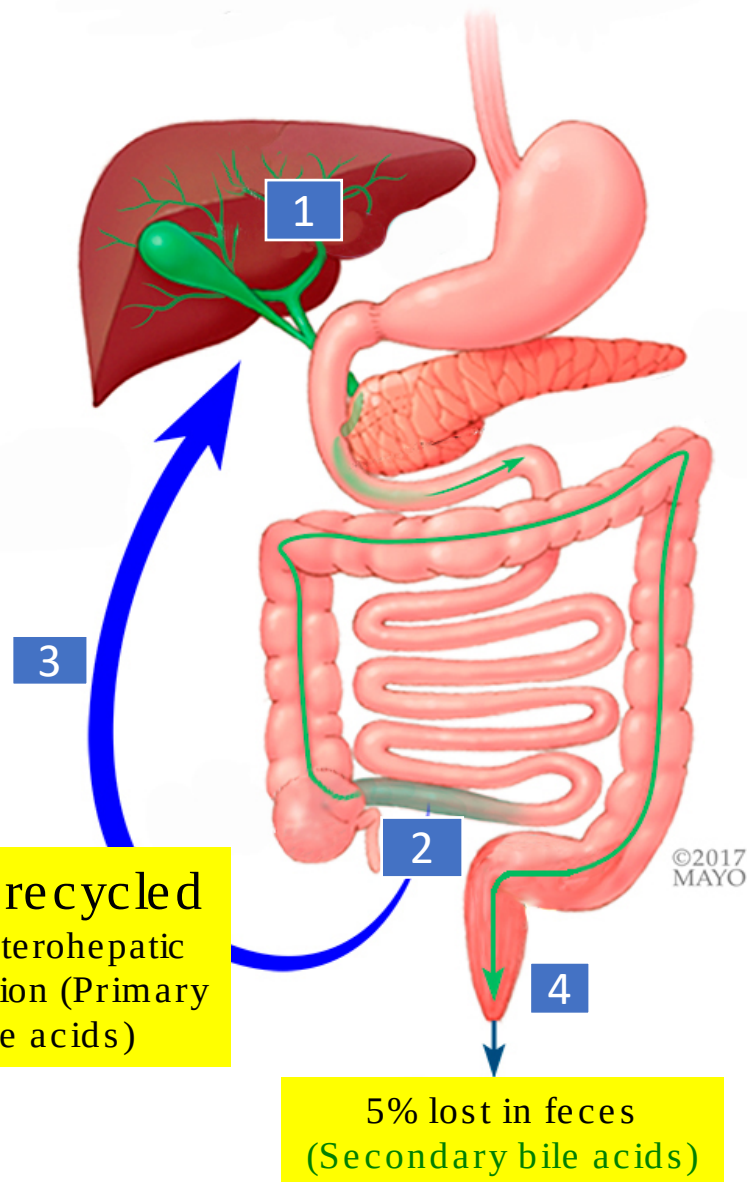
Or Glychenodeoxycholic acid
(Chenodeoxycholic acid + glycine)

Enterohepatic Circulation of Bile Acids/Bile Salts

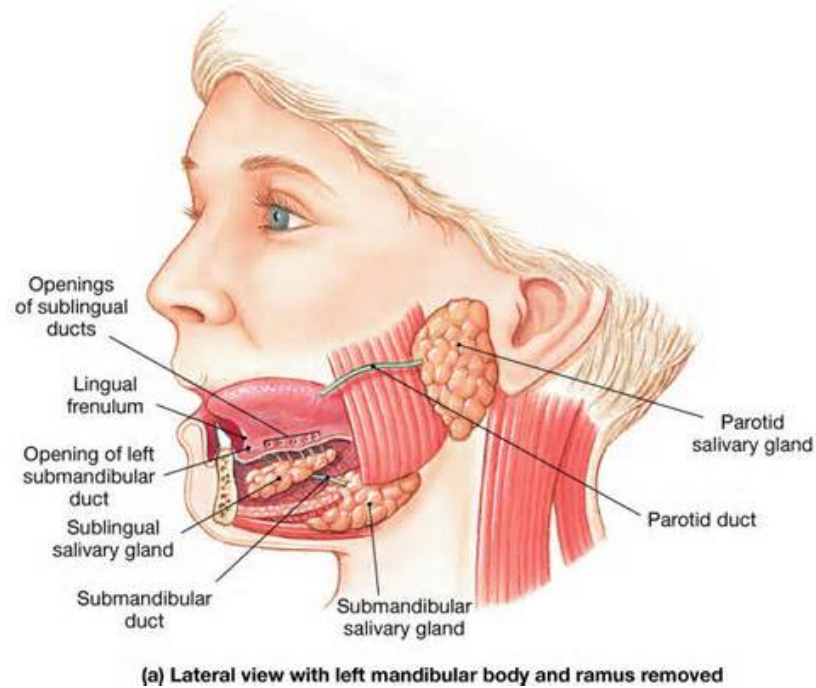


- **Liver conjugates** primary bile acids and **releases conjugated (primary) bile salts** into **bile (1)**
- **Bile salts assist in lipid digestion and absorption** in small intestine
- In **terminal ileum**, most of **primary bile acids** are **absorbed** (95% of bile salts/acids) [2,3]
- **Enterohepatic circulation:** The cycle of bile salt secretion into bile, passage through the duodenum, uptake into the ileum, and subsequent return to liver
- **Intestinal bacteria** convert cholic acid and chenodeoxycholic acid to generate **secondary bile acids/salts** (deoxycholic acid and lithocholic acid respectively) – Most of which is lost in feces [4]

Formation and Enterohepatic Circulation of Bile Salts/Bile Acids



1. Formation of conjugated (primary) bile salts in liver and release into duodenum via bile ducts
2. Reabsorption of 95% of the bile salts/acids from the **terminal ileum**
 - Formation of secondary bile acids by bacteria
3. Transport of primary bile acids back into the liver via the hepatic portal vein (95% recycled) – Conjugated and secreted into bile
4. Loss of bile salts in feces (5% lost in feces) - **Secondary bile acids**



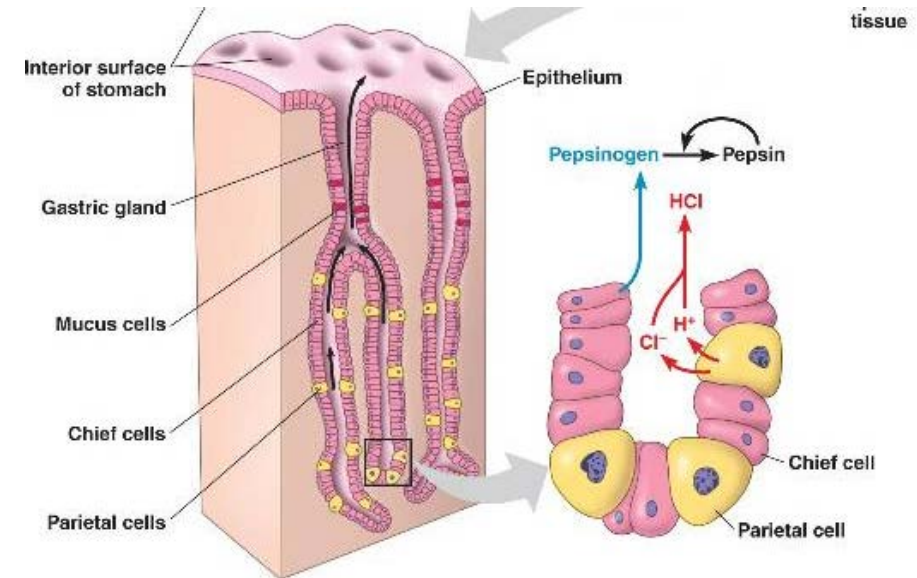
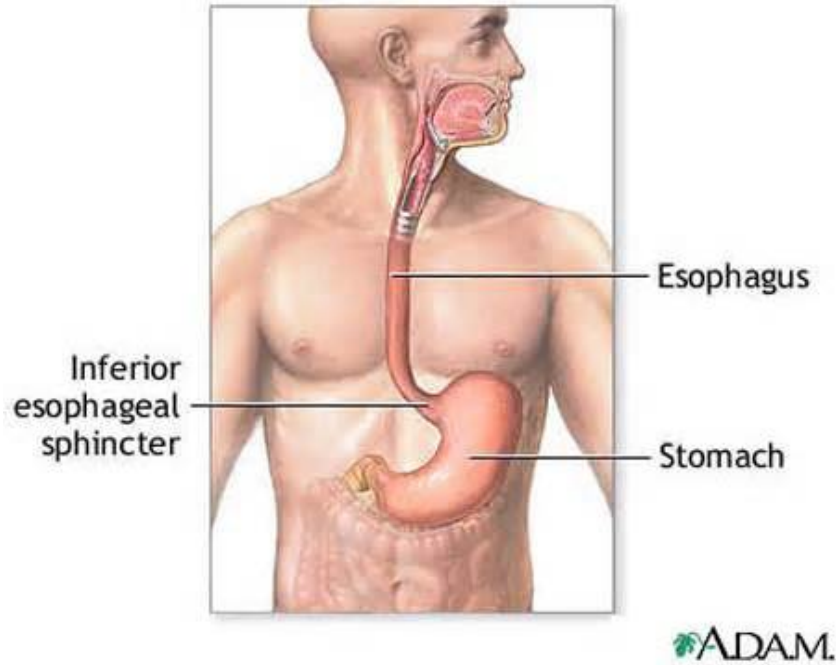
Digestion in the Mouth:

Carbohydrate digestion by salivary alpha-amylase

Salivary amylase cleaves $\alpha(1 \rightarrow 4)$ glycosidic bonds of **starch** and **glycogen** and forms **branched oligosaccharides**.

Lingual lipase for **lipid digestion** is **released** in the **oral cavity** but is not active due to the short time in the mouth

Digestion in the Stomach



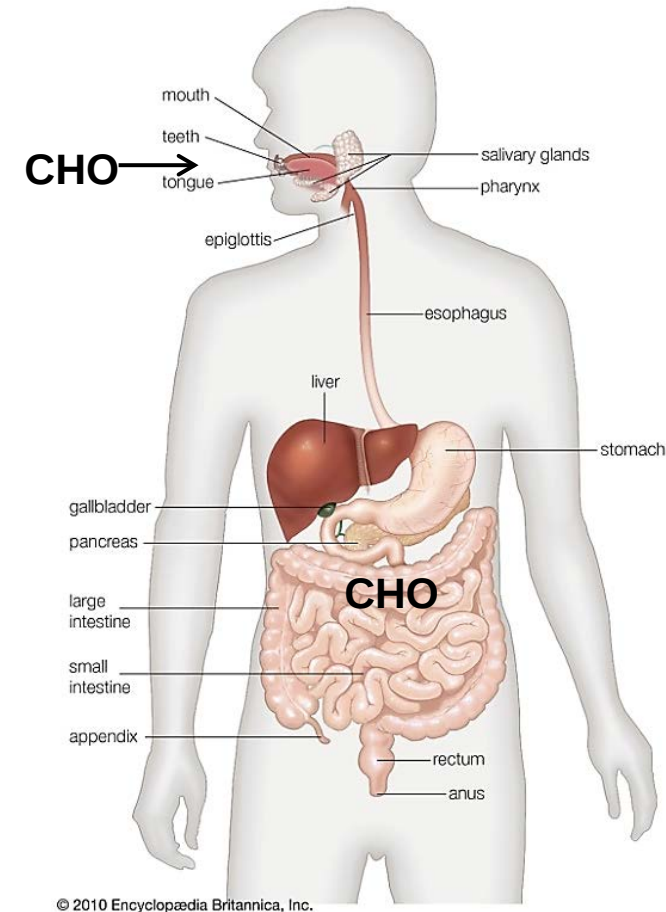
- The **thought** of food, sight, smell, taste, chewing and a **full stomach** stimulates the release of **gastrin** from G-cells of the stomach and duodenum into blood.
- **Gastrin** (peptide hormone) that leads to release of **hydrochloric acid (HCl)** from **parietal cells**.
- The **chief cells** release **pepsinogen** and **gastric lipase**.



Enzyme Activities in Stomach

1. **Carbohydrate digestion**: stops due to **denatured** salivary α -amylase at low pH.
2. **Protein digestion**: starts performed by **pepsin**.
 - Pepsinogen (zymogen) is autocatalytically activated to pepsin at high proton concentration (Low pH due to HCl)
3. **Lipid digestion**: starts with degradation of TAGs with medium-chain fatty acids performed by **lingual** and **gastric lipases** which **degrade** TAGs in milk.
 - **Important for infant nutrition and patients with Cystic Fibrosis (fat malabsorption).**
 - **Not significant in adults**
 - **Milk digestion** generates free **medium-chain fatty acids** which are released into **hepatic portal vein** and directly reach the liver for **energy metabolism**.

Carbohydrates (CHO) digested to glucose, fructose and galactose (monosaccharides)



1. **Mouth:** **salivary α -amylase** degrades starch and glycogen to **dextrins, isomaltose and maltose**.

Humans cannot digest **cellulose** (dietary fiber; lost in feces).

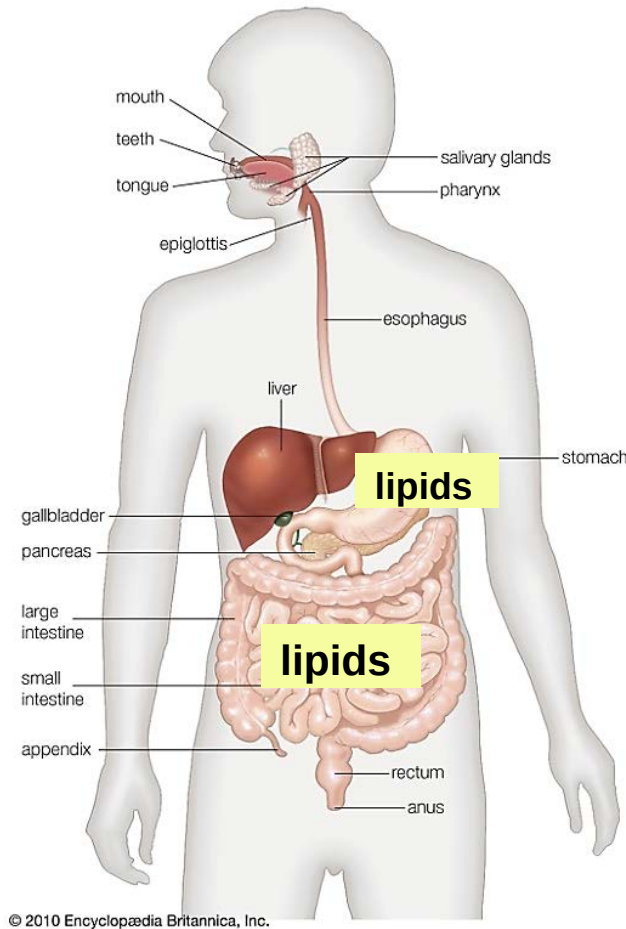
2. **Stomach:** No significant carbohydrate digestion

3. **Small intestine:** **pancreatic amylase** forms **isomaltose and maltose**.

4. **Brush border cells:** **disaccharidases** cleave **lactose, sucrose, isomaltose and maltose** → **Monosaccharides**

Lipid Digestion and Absorption in Small Intestine

- Needs Bile salts



1. Mouth: Lingual lipase is mainly swallowed and active in the stomach.

2. Stomach: Lingual and gastric lipases (acid stable) and act mainly on TAGs with medium-chain fatty acids — **important only in infants**

3. Small intestine: Pancreatic lipase, phospholipase A₂ and cholesteryl esterase need **conjugated bile salts** for activity and dietary lipid emulsification.

4. Brush border cells: Lipid absorption needs mixed micelles, formed by **conjugated bile salts** and lipid digestion products.

Summary Notes – Recommended Reading

Bile Acids/Bile Salts

HA = Protonated → bile acids
A- = Deprotonated (ionized) → bile salts

Synthesised from
Cholesterol in
Hepatocytes



Synthesised from
Primary bile acid in
Hepatocytes



Synthesised from Primary bile salts by
bacteria in the colon and called
Secondary bile acids



Primary bile acid/salt	Conjugated bile salt (Required for lipid digestion and absorption). Also called Primary bile salt; 95% reabsorbed in ILEUM	Secondary bile acid: Primary bile salts are converted into secondary bile acids (by intestinal bacteria). Most of the secondary bile acids are lost in feces
Cholic acid The COOH group will be protonated in ~50% of molecules [bile acids] & deprotonated (ionized) in ~50% of molecules [bile salts].	Conjugated to Glycine to form Glycocholic acid . At alkaline pH (negatively charged) called Glycocholate . Conjugated to Taurine to form Taurocholic acid . At alkaline pH (negatively charged): Taurocholate	Glycocholic acid →→ Deoxycholic acid
Chenodeoxycholic acid	Glychenodeoxycholic acid (Anion is Glychenodeoxycholate) Taurochenodeoxycholic acid (Anion is Taurochenodeoxycholate)	Chenodeoxycholic acid → Lithocholic acid

Thank you

Dr. Felicia Ikolo
fikolo@sgu.edu